



MATHEMATICS

# The Unreasonable Effectiveness of Mathematics in the Natural Sciences

A summary

**Mark Pedigo**

29 Sep 2019 — 6 min read



Werner Heisenberg and Eugene Wigner, 1928

I recently started a job in which I, trained as a mathematician, am working with a number of colleagues who were trained as physicists. At lunch, this situation led to a discussion of the connections between the two fields, which in turn led to a discussion of Eugene Wigner's article "The Unreasonable Effectiveness of Mathematics

Subscribe

in the Natural Sciences.” I reluctantly admitted that although I had heard and read about the article, and thought I understood its general gist, I had never taken the time to read and digest the article for myself. To remedy this sad state of affairs I decided to carefully read it and provide a summary to increase my understanding of the article. I hope that you, the reader, find it helpful as well.

I should note that my intent here is to dissect and digest the article itself, not the responses to it (for instance, Richard Hamming’s 1980 article, “The Unreasonable Effectiveness of Mathematics”) or any questions connected with it. Wigner’s article serves as the tip of a very interesting conceptual iceberg which is concerned with, among other topics, the connections between mathematics and science, and of mathematical philosophy and ontological questions of mathematics. I plan to explore the Wigner’s article here and perhaps address these related issues separately in future articles.

## Eugene Wigner

Eugene Wigner (1902–1995) (natively, Wigner Jenő), attended the Kaiser Wilhelm Institute in Berlin, where he was an assistant to Karl Weissenberg and Richard Becker, who introduced group theory into physics, and then at the University of Göttingen, where he served as assistant to the great David Hilbert. He later worked on the Manhattan Project, after which he hired as the Director of Research and Development at the Clinton Laboratory (now the Oak Ridge National Laboratory). After the war he worked at several governmental institutes, including the National Bureau of Standards, the National Research Council, the National Science Foundation and the General Advisory Committee of the Atomic Energy Commission. In 1960, he received the Nobel Prize in Physics for “contributions to the theory of the atomic nucleus and the elementary particles, particularly through the discovery and application of fundamental symmetry principles.”

## The Unreasonable Effectiveness of Mathematics in the Natural Sciences

In 1960, his article “The Unreasonable Effectiveness of Mathematics in the Natural Sciences” was published in *Communications on Pure and Applied Mathematics*. The article poses two related issues: the effectiveness of mathematics in physics (beyond that of a mere supporting tool), and the uniqueness of mathematical theories in physics. To the first point, the effectiveness of mathematics, he says, “The enormous usefulness of mathematics in the natural sciences is something bordering on the mysterious and there is no rational explanation for it.” To address this issue, he addresses the following questions in turn: “What is mathematics?”, “What is physics?”, “What is the role of mathematics in physical theories?”, and “Is the success of physical theories truly surprising?” To the second point, the uniqueness of physical theories, he has less to say, although his discourse on this subject is also interesting and provocative.

Let's explore his responses to these questions in turn.

## What is mathematics?

Wigner defines mathematics as “the science of skillful operations with concepts and rules invented just for this purpose.” The emphasis is on the concepts, without which, he notes, only a handful of interesting theorems could be proved from the axioms. Furthermore, these concepts are not chosen for simplicity, but rather “with a view of permitting ingenious logical operations which appeal to our aesthetic sense both as operations and also in their results of great generality and simplicity.”

An example of such a mathematical concept is the complex numbers. Complex numbers are not born of experience, but rather of mathematical beauty. The theory of complex numbers, he ascertains, was not pursued for applications, but rather for the beauty inherent in the theory. In other words, mathematicians study mathematics not as ends to a mean, but as a subject in and of itself.

## What is physics?

Physics, on the other hand, is concerned with the laws of nature. Such a law predicts some future event given a present event, or, put another way, is a conditional (if-then) statement. An example is Galileo's law of gravitation — the observation that if two rocks are dropped at the same time from the same height then they will land on the ground below at the same time. Such a law is surprising, asserts Wigner, because it is invariant (the experiment with the rocks does not matter where on Earth you carry it out, or at what time you perform it) and because it depends only on (a) specific variable(s), and not on others that conceivable might have had an effect (the objects need not be rocks, and the result depends only on the mass of the objects, not on size, shape, or other factors one might consider).

## How is math used in physics? Is its success in physical theories truly surprising?

The use of mathematics in physics as a tool is clear. Since physics is formulated in terms of laws written as conditional statements, mathematics is the obvious choice to formulate these expressions. But the role of mathematics in physics goes far beyond this obvious point. According to Wigner, “...the laws of nature must have been formulated in the language of mathematics to be an object for the use of applied mathematics... The mathematical formulation of the physicist's often crude experience leads in an uncanny

number of cases to an amazingly accurate description of a large class of phenomena.” In other words, we “get more out of the equations than we put in.”

He gives three examples of this amazing accuracy: the laws of falling bodies, quantum mechanics, and quantum electrodynamics. Let’s take these examples one at a time.

Isaac Newton noticed that the path of a falling body (perhaps a thrown rock) on the Earth and the path of the moon in the sky are two particular cases of the more general notion of an ellipse. From this observation, he postulated the universal law of gravitation, which states that the gravitation between two objects is proportional to their masses. While Newton, given the restrictions of his day, could only verify the results with an accuracy of 4%, the law was later proved to be accurate to within less than a ten thousandth of one percent. This law, therefore, is a fantastic example of a mathematical formalism that has proved accurate beyond any reasonable expectations.

Quantum mechanics gives an even more astounding example. Matrix algebra had been studied independently of any applications by pure mathematicians for some time when Max Born realized that Werner Heisenberg’s rules of computation were formally identical with the rules of computation of matrices. Born, Pascual Jordan, and Heisenberg then replaced the position and momentum variables of Heisenberg’s equations of classical mechanics by these matrices and applied that result to an idealized problem. The new formulation worked, but would it work in a realistic setting, not just a toy problem? Within months, Wolfgang Pauli applied the new formulation to a realistic problem (a hydrogen atom) and the results matched up. Since Heisenberg’s original calculations were abstracted from problems that included the old theory of hydrogen atoms to begin with, this result was not too surprising. However, the “miracle” occurred next, when the matrix mechanics were applied to problems for which the Heisenberg rules no longer applied. — equations of motion of atoms with greater numbers of atoms. These observations were shown to agree with experimental data to within *one part in ten million*! Once again, mathematics developed independently of physics has been applied to physics to give spectacularly accurate results, far beyond the expectations of the original theory.

The third example is the theory of the “Lamb shift” in quantum electrodynamics. This theory was developed purely in terms of mathematics, devoid of any experimental data, and only tested after the fact. The agreement with experimental data is better than one part in a thousand.

## Uniqueness of physical theories

For the second big issue, the uniqueness of physical theories, Wigner states, “We cannot know whether a theory formulated in terms of mathematical concepts is *uniquely* appropriate.” [emphasis mine]. By this, he means that just because we develop a theory that describes some physical phenomena, we do not know if there

might be another theory, based on for instance, different choices of variables, or different mathematical machinery, that might produce the same results.

For example, consider the incompatibility of the theory of quantum phenomena and the theory of relativity, two models that have their roots in mutually exclusive groups of phenomena. Relativity theory applies to macroscopic bodies (for instance, stars) and is formulated in terms of four-dimensional Riemannian space, whereas quantum theory applies to the microscopic world and is modeled using infinite dimensional Hilbert space. The point of contraction is the so-called “event of coincidence” which refers to the collision of infinitely small particles. The result from relativity theory is primitive and defines a point in space-time, whereas quantum theory yields a non-primitive solution which is not isolated in space-time. We therefore know of an example for which these two theories are directly contradictory. Perhaps there they are both approximations of a larger theory that will explain both cases. As of this writing, we do not know.

## References

Hamming, R. W. (1980). “The Unreasonable Effectiveness of Mathematics”. The American Mathematical Monthly. 87 (2): 81–90.

Wigner, E. P. (1960). “The unreasonable effectiveness of mathematics in the natural sciences. Richard Courant lecture in mathematical sciences delivered at New York University, May 11, 1959”. Communications on Pure and Applied Mathematics. 13: 114.

## Member discussion

0 comments

## Start the conversation

Become a member of **Cantor's Archive** to start commenting.

[Sign up now](#)

Already a member? [Sign in](#)

[READ MORE](#)

